RLC Second-Order Analysis

ENGS152 Circuits Lab

6 March 2026

Step 1: RLC Calculations

We start by finding the personal resonant frequency f_0

$$f_0 = (surname\ length \times name\ length)^2$$

Name length = 5 **Ashot**

Surname length = 10 **Nahapetyan**

$$f_0 = (5 \times 10)^2 = 2500\ Hz$$

Allowed 5% range: 2500 $\pm$ 0.05(2500) = 2500 $\pm$ 125 $\Rightarrow \boldsymbol{[2375\ , 2625]\ Hz}$

Let C = 0.22uF and L = 20mH

$f_0 = \frac{1}{2\pi\sqrt{LC}} \approx \boldsymbol{2399.4\ HZ}$**,** the value is within 5% uncertainty.

Quality factor is **10$\pm$0.5**:

Let **R** = 33 Ω:

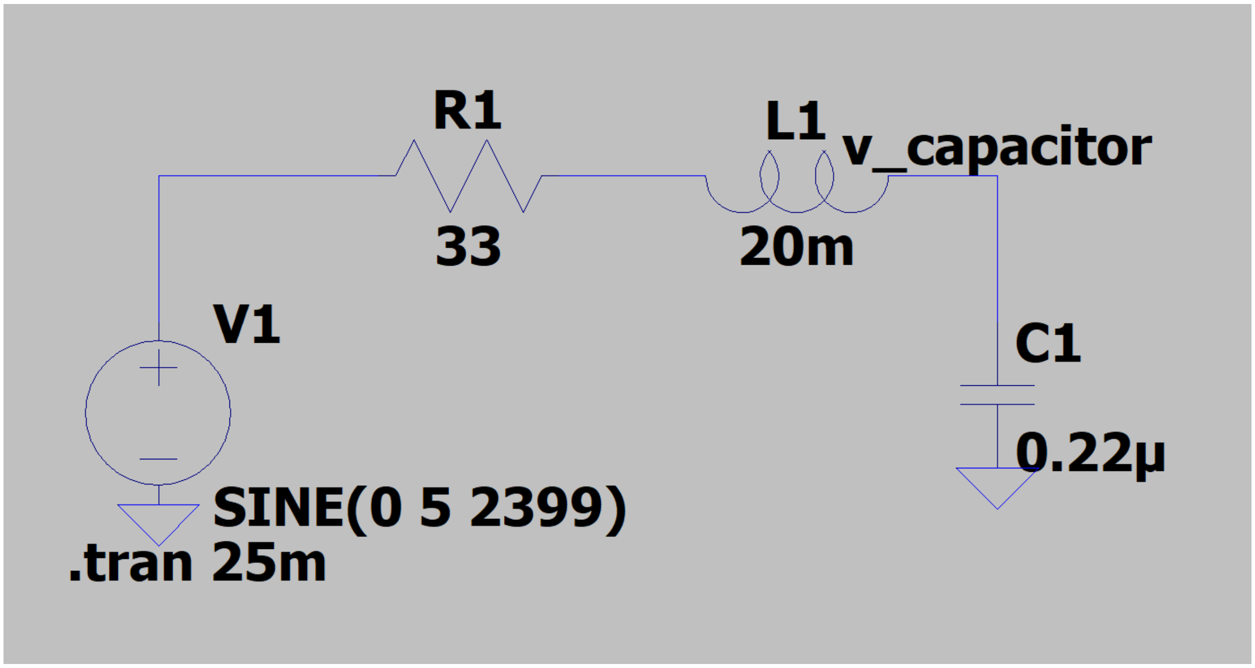
$Q = \frac{1}{R}\sqrt{\frac{L}{c}} \approx 9.14$, the value falls in the range of quality factor Q.

Hence:

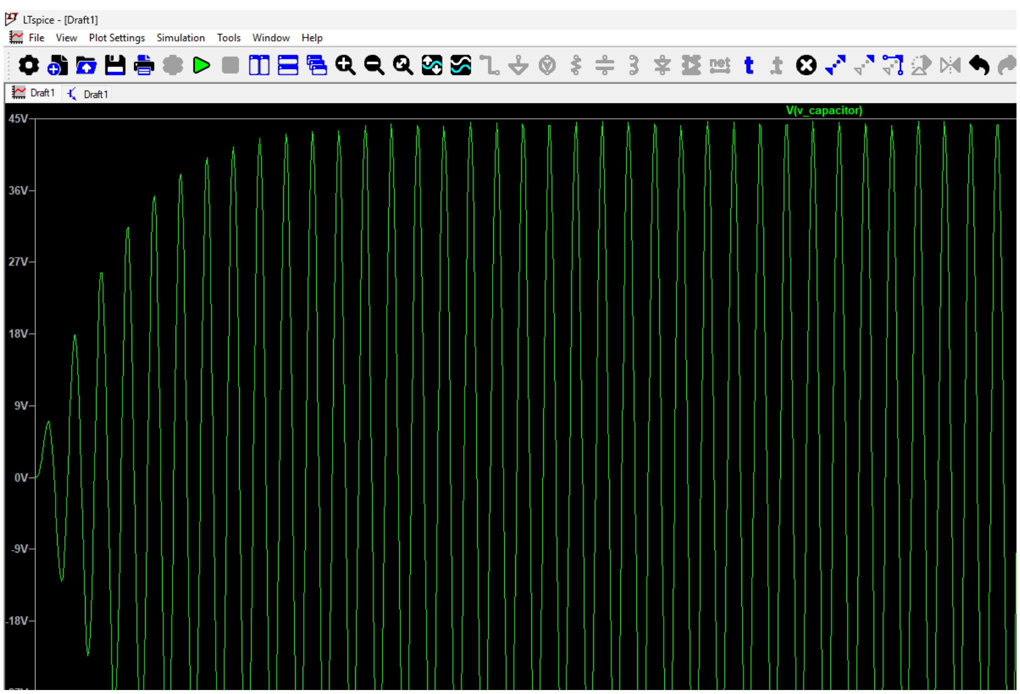
C = 0.22uF, **L** = 20mH, **R** = 33 **Ω**, **Q** = 9.14

Step 2: Circuit Design

First, we are asked to build the RLC circuit in LTspice using the calculated values.



After setting up the voltage source a sine wave with amplitude of 5V and frequency of 2399Hz and running the simulation, it is visible that the voltage on the capacitor reaches value of $V_C \approx 45V$



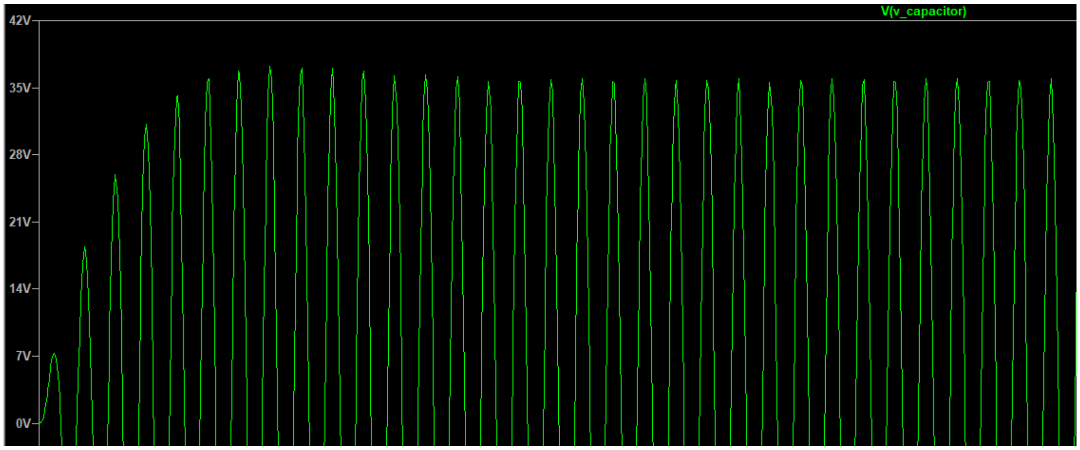
$V_c = \frac{V_{in}}{(1-1^2)^2+\frac{1}{Q}} = V_{in} \times Q$ = 9.14 * 5V ≈ $\mathbf{45.7\ V}$ (This is what happened to the voltage on the capacitor at the resonant frequency) It got amplified by the quality factor.

Bandwidth:

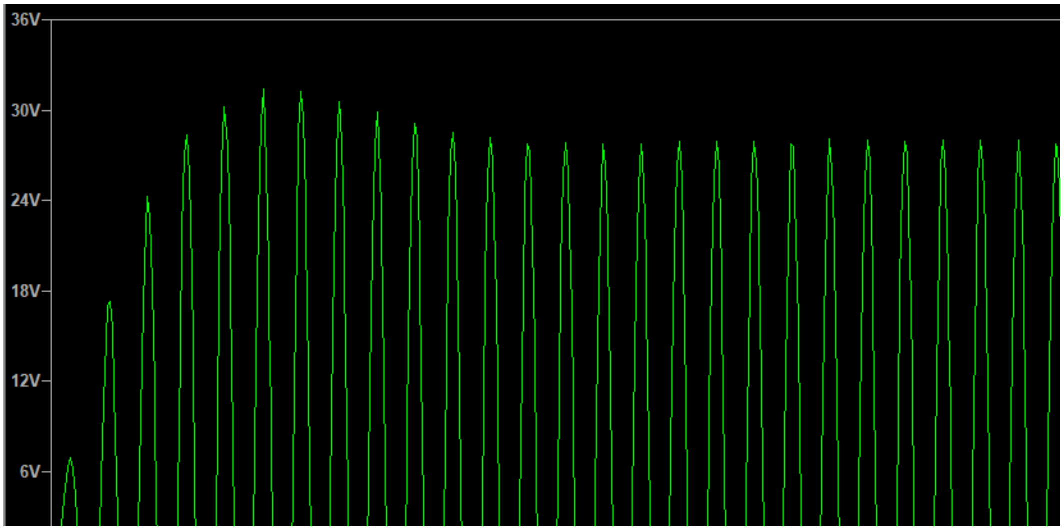
$B = \frac{R}{2\pi L} = \frac{33}{2\pi\times 20\times 10-3} = 262.6\ Hz$ **Bandwidth.** Below you can see the bandwidth edge frequencies.

$f_1 = 2399 - \frac{B}{2} = 2267.7\ Hz$ $f_2 = 2399 + \frac{B}{2} = 2530.3\ Hz$

At $f_1 = 2267.7\ Hz$ Peak to Peak voltage appears to be 38 V.



At $f_2 = 2530.3\ Hz$ Peak to Peak voltage appears to be 32 V.



Lastly, we are asked to find a frequency at which the output voltage is the same as the input voltage.

$$\frac{V_c}{V_{in}} = 1$$

$$1 = \frac{1}{\left(1 - \left(\frac{f}{f_0}\right)^2\right)^2 + \left(\frac{f}{f_0 Q}\right)2}$$

After simplifying and substituting known values we get that $f \approx \mathbf{3380\ Hz} > 2399$ Hz. The graph below confirms it.